# BSTA 550 Syllabus

Key Course Info

- **Late work or a closed Brightspace assignment?** Email your work to **both me and the TA** (addresses on the instructor page).
- **Need an extension?** Email me.
- **Homework:** turn it in any time before the course ends. On-time homework gets feedback; late homework does not.
- **Project checkpoints:** you get ONE no-questions-asked, 3-day extension.
- **Attendance:** try to come to class in-person. Recordings will be available.
- **Used AI on an assignment?** Add an AI Use Statement at the top (see the Generative AI Policy).
- Class ends December 07, 2026. **All coursework MUST be completed by December 11, 2026.**
- Questions or concerns anytime? Use the Ongoing Course Feedback form.

## Description

Welcome to BSTA 550! This course introduces the methods and concepts of probability theory: conditional probability and independence, combinatorics, discrete and continuous random variables, probability distributions, joint distributions, expectation, transformations of random variables, and the central limit theorem. You'll apply these concepts and build statistical reasoning through simulations in R.

### Course Learning Objectives

By the end of this course, you should be able to…

1. Assign probability to a chance event using concepts of probability (including fundamental axioms, properties, and counting).
2. Compute probabilities for *discrete random variables* (including random variables following Bernoulli, binomial, geometric, and Poisson distributions).

3. Compute probabilities for *continuous random variables* (including random variables following Normal, Gamma, and Beta distributions).
4. Perform statistical computations and simulations using R.

## Instructors & Meeting Times

Contact info and office hours are on the instructor page.

| Day | Time | Room |
|---|---|---|
| Mondays | 10:30 AM - 12:00 PM PT | VPT 264 |
| Wednesdays | 10:30 AM - 12:00 PM PT | VPT 264 |

**Known exceptions:**

- 10/12: Async, optional class
- 10/14: Virtual class
- 11/25: No class

## Materials

### Textbook

**An Introduction to Probability and Simulation** by Kevin Ross (free online, updated 8/11/2022)

**Supplemental readings (optional):**

- *Introduction to Probability*, Mark Daniel Ward and Ellen Gundlach, 1st edition (see shared folder for access)
- *Statistical Inference*, Casella and Berger, 2nd edition (the textbook for BSTA 551-552 Math Stat)
- *Introduction to Probability*, Charles M. Grinstead and J. Laurie Snell
- *Probability With Applications and R*, Robert P. Dobrow, Wiley 2013 (eBook available from the OHSU library)
- *An Introduction to R* (free PDF)

### R and RStudio

We'll use R and RStudio (both free) for homework. **Please install them before our second lecture**, and email me if you get stuck. Here's some helpful installation documentation.

1. R: https://www.r-project.org/
2. RStudio: https://posit.co/download/rstudio-desktop/

New to R? I also teach PUBH 523, an introduction to R. You can use its videos, practice, and more on the course site.

### Other helpful resources

- **PennState STAT 414:** course notes for an advanced undergraduate class that overlaps with ours. It's a great source if you'd like to see concepts phrased differently. Not all of our topics are there, but the most important ones are. I don't list matching sections on our schedule because learning to find resources and the right key words is an important skill. If you can't find something, ask me at office hours, after class, or in a private meeting.

### Where things live

- **This website:** all course materials (lecture notes, homework, data examples, recordings).
- **Brightspace:** where you submit assignments, see your grades, and get course announcements.
- **Webex:** virtual office hours. Please stay muted until you want to talk, drop your question (or "I have a question") in the chat so we can go in order, and turn your video on if you're comfortable, since it helps me get to know you and how you learn.

## Course Structure

| Component | Modality | Frequency | Description |
|---|---|---|---|
| Lecture | In person | Twice weekly | Lectures with interactive examples. Sessions are recorded and posted online. **Attendance in-person is encouraged.** |

| Component | Modality | Frequency | Description |
|---|---|---|---|
| Homework | Online | Every 1-2 weeks | 7 assignments to engage with concepts, practice coding, and apply calculations. Working in groups is encouraged, but everyone submits their own work. **Due Sundays at 11 PM** to get feedback, but accepted any time before the course ends. |
| Quizzes | Online | Every 3 weeks | Questions on key concepts, statistical processes, and interpreting output. **Online, open over a few days, and open book.** Work on quizzes independently. |
| Project (Checkpoints + Final Brief) | Online | 4 checkpoints, 1 final brief | Checkpoints guide you through the calculations and background for a real-world scenario, and the final brief summarizes your context and results. Work independently or in pairs. |

## Grading & Requirements

This class uses two types of assessments:

- **Formative:** work meant to help you *learn and practice*. Feedback helps both of us see gaps and celebrate progress.

- **Summative:** work meant to show *how well you've met the learning objectives* and can use your skills outside the classroom.

| Course activity | Type | Due | Percentage of final grade |
|---|---|---|---|
| Homework | Formative | Weekly | 30% |
| Quizzes | Summative | 10/25, 11/08, 12/06 | 24% |
| Project Checkpoints | Formative/Summative | Every 2-3 weeks | 32% |
| Project Final Brief | Summative | 12/11 | 14% |

Letter grades: A ( 93%), A- (90-92%), B+ (88-89%), B (83-87%), B- (80-82%), C+ (78-79%), C (73-77%), C- (70-72%), D (60-69%), F (<60%).

Points are posted to the Brightspace Gradebook ("Grades" in the left navigation).

### How each activity is graded

### Homework (30%)

Everyone has a different amount of time for coursework, and homework is for *learning*, so grading is simple: **you get a check if you complete at least 75% of the question parts, whether your answers are right or wrong.** Each of the 9 homework assignments counts equally, and the TAs grade them for completeness.

- **On time:** you get the check *and* feedback on one or more problems.
- **Late:** you can turn in homework any time before the course ends with no grade penalty, but you won't get feedback. Compare your work to the solutions or bring it to office hours.

Homework is the most helpful work in this class, so I encourage you to keep up with it and put in what effort you can!

### Quizzes (24%)

There are 3 quizzes, each worth 8%. They're graded for correctness, with partial credit for showing your work and reasoning. Quizzes are open book but must be done independently, and AI is not allowed (see the Generative AI Policy). You'll get solutions and feedback afterward, so use them to fill in any gaps before the next quiz.

### Project Checkpoints (32%)

There are 4 checkpoints, each worth an equal share of the 32%. Each one is graded with a rubric that focuses on **accurate calculations** and **clear explanations of your reasoning and background**. You'll get feedback on each checkpoint, and I expect you to use it in your next checkpoint and your final brief. You get **one no-questions-asked, 3-day extension** for a checkpoint (see the late policy below).

**Resubmissions:** checkpoints are meant for learning, so you can revise and resubmit them to earn back points:

| Checkpoint | Points back on resubmission | Resubmission deadline |
|---|---|---|
| Checkpoints 0 and 1 | **Full credit** back | Checkpoint 0: we'll set aside class time to work on resubmissions. Checkpoint 1: resubmit on Brightspace by **11/4** |
| Checkpoints 2 and 3 | **Half credit** back | Resubmit on Brightspace **within 1 week** of when feedback is published |

For example, if you lose 4 points on Checkpoint 2 and fix everything in your resubmission, you'll earn back 2 points. Resubmit the same way as the original: on Brightspace, as a rendered HTML file plus your `.qmd` file.

### Project Final Brief (14%)

The final brief brings together your checkpoint work into a summary of your scenario. It's graded with a rubric that looks at **accurate results**, **clear context for your scenario**, and **communicating your findings to your audience**. Because it's due at the end of the quarter, feedback will be lighter, but feel free to email me with questions about your grade.

## Course Policies

### Attendance

You're expected to attend class, take part in in-class polls, and complete the exit ticket. **Please try attend classes in person.** Attendance is not part of your grade and recording will be available. I cannot gaurentee recordings will be perfect, sometimes there are technical issues! In-person attendance is your best bet to get the class materials and create community with your classmates!

Exit tickets are meant for points of confusion and clarification!

If a religious holiday falls on a class day, please email me and I'll excuse the absence. Recordings will be available, and you're welcome to bring questions to office hours.

### Late Work Policy

I encourage you to submit work on time, but I know life happens. Here are some examples of extenuating circumstances (not a complete list, so please reach out with questions).

- **Homework:** turn it in any time before the course ends for full credit. For feedback, turn it in on time. **Staying on track with homework really matters**, since falling behind only adds stress to the next one!
- **Project checkpoints:** you get **ONE no-questions-asked, 3-day extension**. Use it wisely! Just email me: "I am using my no-questions-asked extension for Checkpoint ___." After that, email me to explain your circumstances and ask for an extension.
- **Late submissions:** if an assignment is closed on Brightspace, email your work to **both me and the TA**.
- **All coursework must be completed by December 11, 2026 at 11 PM.** If extenuating circumstances mean you need more time, or may affect your work for several weeks, contact me and we'll make a plan together (including any registrar logistics).

If you have an emergency

Whether it involves yourself, family, a pet, a friend, a classmate, or anyone/anything important to you, **please don't worry about contacting me right away.** We'll work it out afterward. If I reach out during an emergency, it's only because I'm worried, and you do NOT need to respond until you're able.

### Regrade Policy

If you think something was graded incorrectly, first compare your answer to the answer key. If you still think a regrade is warranted, email me the question and a short explanation **within one week** of the assignment being returned.

### Generative AI Policy

*This policy is adapted from the OHSU-PSU SPH Biostatistics program's AI policy, which classifies BSTA 550 as a **foundational** course. Everything you need for this class is below.*

**Using AI is *not* required in this class.** If you choose to use it, please follow these guidelines. AI tools (like ChatGPT, Claude, Copilot, Gemini, or an AI assistant built into RStudio) are

becoming an essential skill for biostatisticians, so the goal is to use them *responsibly*, not avoid them.

The toolbox analogy

Think of this class as assembling a toolbox. A new handyperson might need help finding their tools or remembering which one does what, but eventually using them becomes second nature.

**For now, generative AI can help us find and use our tools, but we're working toward using them as second nature!**

### Four guiding principles

1. **Learning comes first.** Ask yourself: *Is AI helping me learn this skill, or doing the skill for me?* If it's doing the skill the assignment is meant to teach, don't use it that way.
2. **Understanding is the priority.** You should be able to explain, reproduce, and defend (without AI) any code, calculation, or conclusion you turn in. Following along with an AI answer is not the same as understanding it!
3. **Build the foundation first.** In this course, working through probability calculations and writing R code *is* the skill we're practicing. That's why some AI uses are limited here, not because AI is untrustworthy.
4. **Be honest about AI use.** Tell me when and how you used AI (see the AI Use Statement below).

### What's allowed

*Quizzes means while you're actively taking a quiz, not while studying for one.*

| **Can I use AI to...** | **Homework** | **Project** | **Quizzes** |
|---|---|---|---|
| ***Coding*** | | | |
| Explain an error message | Yes | Yes | No |
| Explain what a function does | Yes | Yes | No |
| Help debug my code | Yes | Yes | No |
| Give me extra practice problems | Yes | Yes | No |
| Style or tidy my code | Yes | Yes | No |
| Check an answer I already calculated by hand or by code | Yes | Yes | No |

| **Can I use AI to...** | **Homework** | **Project** | **Quizzes** |
|---|---|---|---|
| Draft code for repetitive data management tasks | No | Yes | No |
| Draft code for an analysis or explore approaches beyond what we've learned | No | No | No |
| Take an assignment question and write the code for me | No | No | No |
| Write code to calculate an answer we're learning to do by hand | No | No | No |
| ***Probability and statistics*** | | | |
| Explain a concept or method in general terms (not applied to my assignment) | Yes | Yes | No |
| Check reasoning I've already written about which distribution or method to use | Yes | Yes | No |
| Pick the distribution or method for me | No | No | No |
| Show the work (by hand or code) for a specific problem | No | No | No |
| Interpret my results for me | No | No | No |
| ***Writing*** | | | |
| Give grammar or clarity feedback on my own draft | Yes | Yes | No |
| Brainstorm structure or an outline | Yes | Yes | No |
| Critique an interpretation I wrote myself | Yes | Yes | No |

| Can I use AI to... | Homework | Project | Quizzes |
|---|---|---|---|
| Help find background sources (that I then read and verify) | Yes | Yes | No |
| Draft my background or interpretation of results | No | No | No |
| Paste in AI-written text without substantial editing | No | No | No |

There may be exceptions. For example, if a question asks *you* to explain what a function does, you can't paste in AI's explanation. When in doubt, just ask me!

Never allowed

- Copying and pasting AI-generated answers into your assignment
- Pasting all or part of an assignment into AI and asking for the solution
- Presenting AI output as your own reasoning (e.g., asking AI what to conclude and writing it as your own conclusion)
- Using AI during a quiz
- Uploading confidential or restricted data (including any data with protected health information) into an AI tool, unless it's a university-approved platform under a data use agreement
- Using AI-generated datasets or output as if they came from the assigned data and methods
- Using AI-generated references, results, or facts without verifying them yourself
- Submitting AI-assisted work without an AI Use Statement

Even for allowed uses, **always inspect AI-generated code and its output**, whether it's a whole chunk or a single comma. You're responsible for making sure it does what it's supposed to.

Homework is graded on completion, so I'd much rather see a half-written answer that shows your thinking than a polished answer from AI. And if AI helped you understand something, tell me! I'd love to incorporate it into future classes.

### AI Use Statement

If you used AI on an assignment, add a brief statement at the top that includes **which tool(s)** you used, **what for** (a sentence or two), and **where** it contributed to any written interpretations. For example:

> *AI Use Statement: Claude was used to explain an error message in Q3 and to give grammar feedback on my final brief. All statistical reasoning, code review, and conclusions are my own.*

You don't need to share a link or transcript of your conversation.

### If the guidelines aren't followed

If I or a TA notice AI was used outside these guidelines, we'll reach out to let you know which assignment and which guideline, and you can choose to:

a. Take a 0 on the assignment as submitted,
b. Redo the assignment without AI for a new score, or
c. (Code and hand calculations only) Explain to me (by email or in a meeting) exactly how you used AI and show that you can do the work on your own. If that works out, you keep your original grade; if not, the question(s) where AI was used get 0 points.

First-time, good-faith mistakes (like forgetting an AI Use Statement) will generally be treated as a learning moment, not an integrity issue. Undisclosed AI use, using AI on a quiz, or turning in work you can't explain may lead to consequences under this policy and university academic conduct policies.

### Academic Integrity

As a PSU/OHSU student, you're expected to act with professionalism and integrity. Academic misconduct (plagiarism, cheating, fabricating data, falsifying records, or helping others do so) will be taken seriously. Quizzes must reflect your own individual effort, and any resources you use should be cited.

## How to Succeed in This Course

Every instructor uses resources and assignments a little differently, so here's how to get the most out of each in *this class*.

### Resources

| Resource | How to use it |
|---|---|
| Lectures and recordings | Come in person if you can. Follow the slides, take notes on details mentioned out loud, work through examples with me, and ask questions whenever something isn't clear. |
| Office hours | Ask questions about assignments or lecture notes, work on homework with help nearby, or meet classmates to collaborate with. |
| Textbooks | Not required reading. Reading ahead can help during class, but I see the textbook as a resource for when you're stuck on a topic, need an example for homework, or want more practice before a quiz. |
| Website | Home for all course materials. Each week's page on the schedule has its lecture notes, homework, and recordings. Explore the tabs to get a sense of how it's organized. |

### Assignments

| Assignment | Before you submit | After it's graded |
|---|---|---|
| Homework | • Work out each problem on your own as much as you can • Talk through problems with a peer • Come to office hours for help • Show your thought process • Search Stack Overflow for R errors | • Review the solutions and your mistakes • For written answers, check with me or a TA if yours is sufficient • Bring questions to office hours |
| Quizzes | • Identify the learning objectives in each lecture • Understand *why* certain tools are used for certain cases • Practice testing yourself and others • Bring specific problems to office hours | • Review the solutions and your mistakes • For written answers, check with me or a TA if yours fits the solution • Only ask for a regrade after viewing the solutions |
| Project | • Start each checkpoint early • Check your calculations with a partner or classmates • Bring specific questions to office hours | • Use checkpoint feedback in your next checkpoint and final brief • Email me with grade questions |

Want another part of the course explained like this? Let me know through the Ongoing Course Feedback form.

## Course Expectations

### What you can expect from me

- **Commitment to your success.** I believe everyone can succeed in this course, and I've designed it to support your learning. Talk to me before or after class, or stop by my office, about anything that's unclear.
- **An inclusive, supportive community.** I'll work to build a space of generous listening, civility, humility, and patience, where we examine content from multiple perspectives and you feel free to share your views and ask questions.
- **Openness to feedback.** Tell me when something isn't clear or isn't meeting your needs, and especially if you notice bias or stereotyping in my materials. Tell me what you enjoy, too!
- **Responsiveness.** I check email daily and aim to reply within 24 hours, Monday-Friday.
- **Clear guidelines and prompt feedback.** Clear instructions and rubrics for assignments. The TAs and I will give helpful feedback and update grades promptly in Brightspace.

### What I ask of you

- **Attend class** (at least 12 classes in person) and complete each exit ticket.
- **Participate** in class and office hours, and be respectful of everyone's contributions.
- **Reach out early.** If you're having trouble keeping up, let me know as soon as possible so we can find a solution together.
- **Submit assignments** through Brightspace, following the late policy if you need more time.
- **Tell me about accommodations** before or during the first week. Students seeking academic accommodations should register with the appropriate service (see the policies on Brightspace).
- **Take care of yourself** and seek help if you need it (see Student Wellness below).

## Course Communications

- **Brightspace announcements:** for important or urgent updates, I'll post on Brightspace (sent to your OHSU email).
- **Course questions:** ask about assignments, materials, or assessments in class or at office hours, or email me. Please give me 24 hours to respond, Monday-Friday. Work-life balance matters for me too!

- **Email:** please use my OHSU address (wakim@ohsu.edu). **Messages sent through Brightspace Inbox will not be answered.**

## Course Feedback

### Ongoing Course Feedback

You can share anonymous feedback anytime through this Microsoft Form or the class exit tickets (leave your email only if you'd like a reply). I review it regularly. If I've done anything that made you uncomfortable, please tell me so I can change. **Ultimately, this class is for you, and my social identity/behavior should not get in the way of your learning.** Some examples:

- Nicky talks a little fast during lecture. Could you slow down?
- During office hours, Dr. Wakim made a face when I asked a question, and it made me feel self-conscious.
- Dr. W asked about my experience in a way that made me feel like a monolith. Please don't assume I can speak for my social identity groups.
- The in-class examples don't make me more interested in the material.

### Midterm Feedback

In the middle of the quarter, I'll ask for guided, anonymous feedback. It really helps me adjust the class while you're still in it!

### Final Course Feedback

At the end of the course, you'll be asked to complete a formal evaluation. **Your feedback is critical for improving this course and matters for my career advancement.** Since our class is on the smaller side, everyone's participation is needed for the feedback to be released. Thank you!

## Schedule

See the Schedule page. I may adjust it if we need more or less time on a topic. You don't need to read the textbook chapters before each class.

## Student Resources

### Academic Success Center

OHSU's Academic Success Center helps all learners find the resources and community they need, including learning skills support, writing support, English for speakers of other languages (ESOL) support, and individual and group tutoring.

### Student Wellness

I care about your physical and emotional well-being. If a health issue is affecting your work, or you'd like help connecting with resources, please reach out.

- **OHSU:** Behavioral Health (**all visits are free**) and Student Health and Wellness, 503-494-8665
- **PSU:** Student Health and Counseling (SHAC), 503-725-2800. OHSU students can also use PSU's Counseling Services.
- SPH Health and Wellness Resources

### Food Insecurity

- PSU's Free Food Market, second Monday of each month
- SNAP for college students: SNAP provides money for food. If you're a single person making less than $2,609/month, you may be eligible. The state's Oregon food benefits page has more on how to apply.

### Students with Children

PSU's Resource Center for Students with Children offers family-friendly study spaces, baby starter packs, free kids' clothing, and childcare financial resources.

## SPH and OHSU Policies

See the Syllabus tab in Brightspace for school policies and resources, OHSU competencies, and institutional policies. I don't include them here because I don't have the bandwidth to keep all those links up to date.